Literature Status: Fractional Sobolev Liftings over Compact Universal Covers

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Status: literature already answered. The finite-fundamental-group lifting question in Section 1.3 of arXiv:1709.08565 is completely answered by Theorem 1 of the separate paper arXiv:1907.01373. For compact nontrivial covers, every fractional Sobolev map lifts exactly outside the exceptional interval $1 \leq sp < 2$.

1 Original question

Monteil and Van Schaftingen ask in Section 1.3, arXiv PDF page 4, whether the condition $sp \notin [1, m)$ is necessary for the lifting property when $0 < s < 1$ and the compact target $\mathcal{N}$ has a nontrivial finite fundamental group [1]. The surrounding discussion concerns the universal covering and a simply-connected domain manifold $\mathcal{M}$.

Equivalently, the unresolved range included

$$2 \leq sp < m,$$

where the target's universal cover is compact because $\pi_1(\mathcal{N})$ is finite.

approximation in $W^{1,3}(\mathcal{M}, \mathbb{S}^2)$ [9].

1.3. **Lifting.** Another situation in which Sobolev maps enjoy a uniform bound principles is the lifting problem. Given a manifold $\mathcal{F}$ and a Lipschitz map $\pi : \mathcal{F} \to \mathcal{N}$, it can be checked immediately that if $\varphi \in W^{s,p}(\mathcal{M}, \mathcal{F})$, then $\pi \circ \varphi \in W^{s,p}(\mathcal{M}, \mathcal{N})$. The lifting problem asks whether every map $u \in W^{s,p}(\mathcal{M}, \mathcal{N})$ can be lifted to a map $\varphi \in W^{s,p}(\mathcal{M}, \mathcal{F})$ such that $\pi \circ \varphi = u$ on $\mathcal{M}$, that is, there exists φ such that the diagram

$$\begin{tikzcd} \mathcal{M} \arrow[r, "u"] \arrow[d, "\varphi"'] & \mathcal{N} \\ \mathcal{F} \arrow[ur, "\pi"'] & \end{tikzcd}$$

commutes. In other words, we wonder whether the composition operator $\varphi \in W^{s,p}(\mathcal{M}, \mathcal{F}) \to \pi \circ \varphi \in W^{s,p}(\mathcal{M}, \mathcal{N})$ is surjective.

This lifting problem has been the object of a detailed study when $\mathcal{N}$ is the unit circle $\mathbb{S}^1$ and $\pi : \mathbb{R} \to \mathbb{S}^1$ is its *universal covering*, defined by $\pi(t) = (\cos t, \sin t)$ for every $t \in \mathbb{R}$. In this case, when the manifold $\mathcal{M}$ is simply-connected, every map in $W^{s,p}(\mathcal{M}, \mathbb{S}^1)$ admits a lifting if and only if either $s = 1$ and $p \geq 2$, or $s < 1$ and $sp < 1$, or $s < 1$ and $sp \geq m$ [16]. Similar results hold for the universal covering $\pi : \mathcal{F} \to \mathcal{N}$ when the fundamental group $\pi_1(\mathcal{N})$ is infinite [11]; when $\pi_1(\mathcal{N})$ is a nontrivial finite group, it is not yet known whether the condition $sp \notin [1, m)$ is necessary when $s < 1$. These results apply to the case of the universal covering of the projective space $\mathbb{R}P^m$ by the sphere $\mathbb{S}^m$ when $m \geq 2$ [4, 52].

Another lifting problem that has been studied is the *lifting problem for fibrations*. For the Hopf fibration $\pi : \mathbb{S}^3 \to \mathbb{S}^2$, in contrast with the universal covering, some gauge invariance property shows that the existence of one lifting implies the presence of a continuum of liftings and a lifting is known to exist when $s = 1$ and $1 \leq p < 2 \leq m$ or $p \geq m \geq 3$ or $p > m = 2$ [11], and known to be impossible for some map if $2 \leq p < m$ [9, 11].

To quantify the lifting of a Sobolev map we define the *lifting energy* of a map $u : \mathcal{M} \to \mathcal{N}$ by

$$\mathcal{E}^{\mathrm{lift}}_{s,p}(u, \mathcal{M}) := \inf \left\{ \mathcal{E}_{s,p}(\varphi, \mathcal{M}) \ : \ \varphi : \mathcal{M} \to \mathcal{F} \text{ is measurable and } \pi \circ \varphi = u \right\}.$$

When $s = 1$ and $p \geq 1$, the lifting $W^{1,p}(\mathcal{M}, \mathbb{S}^1)$ preserves the Sobolev energy; when $s < 1$ and $sp < 1$, the existing bounds on liftings of maps in $W^{s,p}(\mathcal{M}, \mathbb{S}^1)$ are linear [16] (see also [51]) and suggest the following uniform boundedness principle:

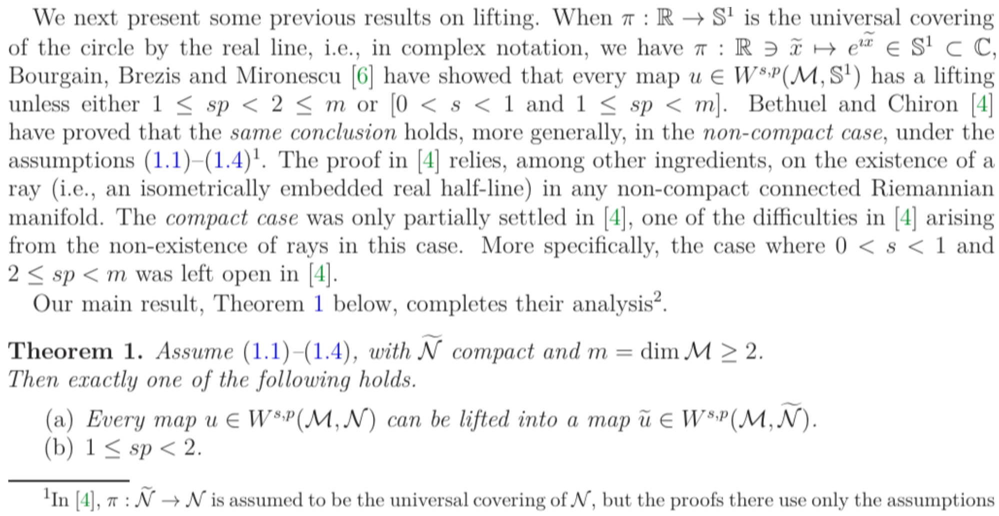
Figure 1: Source Section 1.3, arXiv PDF page 4.

2 Separate later answer

Mironescu and Van Schaftingen subsequently proved a complete compact-cover classification in *Lifting in compact covering spaces for fractional Sobolev mappings*, arXiv:1907.01373 [2]. Their assumptions are a nontrivial Riemannian covering $\pi : \widetilde{\mathcal{N}} \to \mathcal{N}$ with both manifolds compact and connected, and an m-dimensional compact simply-connected domain $\mathcal{M}$, possibly with boundary.

Separate later theorem. *For $0 < s < 1$ and $m \geq 2$, every map $u \in W^{s,p}(\mathcal{M}, \mathcal{N})$ has a lifting $\widetilde{u} \in W^{s,p}(\mathcal{M}, \widetilde{\mathcal{N}})$ if and only if $sp \notin [1, 2)$.*

This is Theorem 1 on arXiv PDF page 2. Its positive part includes the full range $2 \leq sp < m$ that remained uncertain in the source. Its negative part states that not every map lifts when $1 \leq sp < 2$. The abstract explicitly says that the paper settles completely the Sobolev lifting question over covering spaces.



We next present some previous results on lifting. When $\pi : \mathbb{R} \to \mathbb{S}^1$ is the universal covering of the circle by the real line, i.e., in complex notation, we have $\pi : \mathbb{R} \ni \widetilde{x} \mapsto e^{\imath\widetilde{x}} \in \mathbb{S}^1 \subset \mathbb{C}$, Bourgain, Brezis and Mironescu [6] have showed that every map $u \in W^{s,p}(\mathcal{M}, \mathbb{S}^1)$ has a lifting unless either $1 \le sp < 2 \le m$ or $[0 < s < 1$ and $1 \le sp < m]$. Bethuel and Chiron [4] have proved that the *same conclusion* holds, more generally, in the *non-compact case*, under the assumptions (1.1)–(1.4)[1]. The proof in [4] relies, among other ingredients, on the existence of a ray (i.e., an isometrically embedded real half-line) in any non-compact connected Riemannian manifold. The *compact case* was only partially settled in [4], one of the difficulties in [4] arising from the non-existence of rays in this case. More specifically, the case where $0 < s < 1$ and $2 \le sp < m$ was left open in [4].

Our main result, Theorem 1 below, completes their analysis[2].

Theorem 1. *Assume (1.1)–(1.4), with $\widetilde{\mathcal{N}}$ compact and $m = \dim \mathcal{M} \ge 2$. Then exactly one of the following holds.*

(a) *Every map $u \in W^{s,p}(\mathcal{M}, \mathcal{N})$ can be lifted into a map $\widetilde{u} \in W^{s,p}(\mathcal{M}, \widetilde{\mathcal{N}})$.*
(b) $1 \le sp < 2$.

[1]In [4], $\pi : \widetilde{\mathcal{N}} \to \mathcal{N}$ is assumed to be the universal covering of $\mathcal{N}$, but the proofs there use only the assumptions

Figure 2: Supporting Theorem 1, arXiv PDF page 2.

3 Identification and scope

If $\mathcal{N}$ is compact and $\pi_1(\mathcal{N})$ is finite, its universal cover is a finite-sheeted cover of a compact space and is therefore compact. Hence the later theorem applies directly to the source question.

For $m \ge 3$, the answer to the proposed necessity is negative: maps lift for every $2 \le sp < m$, even though these exponents lie in $[1, m)$. The correct exceptional interval is $[1, 2)$. When $m = 2$, the intervals $[1, 2)$ and $[1, m)$ coincide. The packet does not assert anything about nonsimply-connected domains or cases outside the hypotheses of the covering theorem.

This is an exact literature answer, not an original result. The answering paper cites the source paper, shares Jean Van Schaftingen as an author, and explicitly presents its theorem as completing the covering-space lifting problem.

4 Search and verification status

A bounded search through 9 August 2026 checked the run registry, the local arXiv corpus, exact question wording, compact-cover terminology, and later lifting papers. Both source PDFs were rendered, and the decisive passages were inspected at their stated pages. The exact match is arXiv:1907.01373 and its published Analysis & PDE version.

References

[1] A. Monteil and J. Van Schaftingen, *Uniform boundedness principles for Sobolev maps into manifolds*, Ann. Inst. H. Poincaré C Anal. Non Linéaire **36** (2019), 417–449; arXiv:1709.08565.

[2] P. Mironescu and J. Van Schaftingen, *Lifting in compact covering spaces for fractional Sobolev mappings*, Analysis & PDE **14** (2021), 1851–1871; arXiv:1907.01373; DOI 10.2140/apde.2021.14.1851.